

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2025 4WM1H Worked Solutions

Nov2025 | 4WM1H | Unit 1

37 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 2****TOPIC: Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 1 The diagram shows a semicircle.

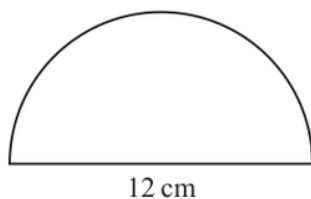


Diagram **NOT**
accurately drawn

The semicircle has diameter 12 cm

Work out the perimeter of the semicircle.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 1 is 2 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 2

TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

(Total for Question 2 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

3 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 3 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 3

Marks: 3

TOPIC: **Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4** **Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

4 $\mathcal{E} = \{8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18\}$

$A = \{\text{odd numbers}\}$

$B = \{\text{multiples of 3}\}$

$C = \{\text{prime numbers}\}$

(a) List the members of the set $A \cup C$
(1)(b) List the members of the set $(A \cap B)'$
(1)(c) Is it true that $15 \in (B \cap C)$?

Tick (✓) the correct box.

Yes

☐

No

☐

Give a reason for your answer.

.....
(1) D is a set such that $D \cup C = A$ D has 2 members.(d) List the members of D
(1)**(Total for Question 4 is 4 marks)**

PAST PAPER SOLUTION

Q#: 4

Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION

Q#: 5 Marks: 5

TOPIC: **Probability Toolkit**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 5 All students at a school are asked which sport they prefer from football, tennis, rugby and cricket.

The table gives information about the probabilities that, when a student is chosen at random, the student prefers football or tennis or rugby or cricket.

Favourite sport	football	tennis	rugby	cricket
Probability	$2x$	0.2	x	0.35

One of the students at the school is chosen at random.

- (a) Work out the probability that this student prefers tennis or cricket.

.....
(1)

There are 700 students at the school.

- (b) Work out how many students prefer football.

.....
(4)

.....
(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION

Q#: 5

Marks: 5

TOPIC: **Probability Toolkit**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6** **Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

6 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g
(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm
(1)

(Total for Question 6 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 6

Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 7****Marks: 2**TOPIC: **Algebraic Roots & Indices**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

7 $\frac{8^{-2} \times 8^9}{8^{10}} = 8^n$

Work out the value of n $n = \dots\dots\dots$

(Total for Question 7 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 7

Marks: 2

TOPIC: **Algebraic Roots & Indices**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 9****TOPIC: Factorising**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

8 (a) Expand $5w(3w - 4w^2)$
(2)(b) Simplify fully $(4a^3c^5)^3$
(2)(c) Factorise fully $14m^3p^2 + 21m^2p^4$
(2)(d) (i) Factorise $x^2 - 10x + 24$
(2)(ii) Hence solve $x^2 - 10x + 24 = 0$
(1)

(Total for Question 8 is 9 marks)

PAST PAPER SOLUTION

Q#: 8

Marks: 9

TOPIC: **Factorising**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 9****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 9 The diagram shows a sketch of a right-angled triangle, ABC

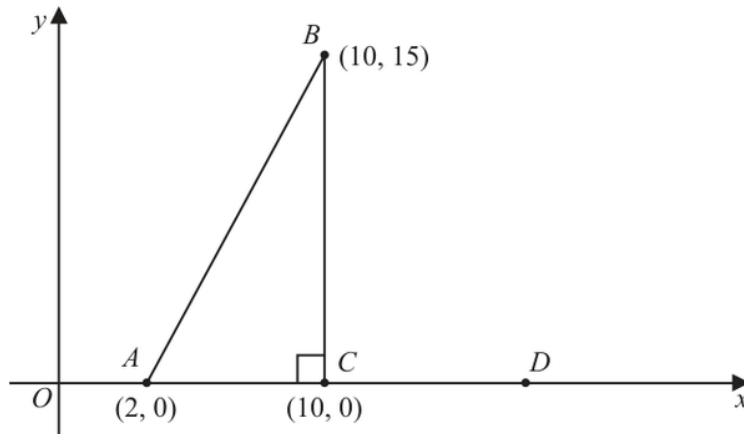


Diagram **NOT**
accurately drawn

A is the point with coordinates $(2, 0)$

B is the point with coordinates $(10, 15)$

C is the point with coordinates $(10, 0)$

D is a point on the x -axis such that $AB = AD$

Find the coordinates of D

(.....,)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Standard & Compound Units**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 10** A parcel in the shape of a cube is placed on a table so that the whole of one face of the parcel is in contact with the table.

The parcel has sides of length 7 cm

The pressure on the table due to the parcel is 62 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the force exerted by the parcel on the table.

..... newtons

(Total for Question 10 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 10

Marks: 3

TOPIC: **Standard & Compound Units**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

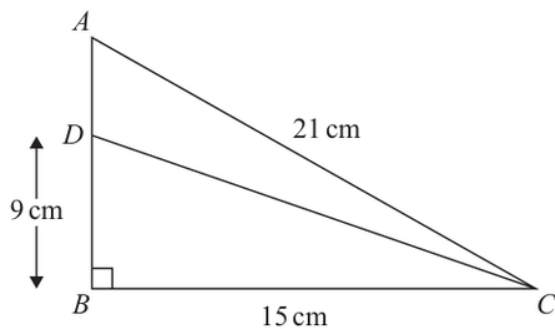
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

11 ABC is a right-angled triangle.Diagram **NOT**
accurately drawn

$$AC = 21 \text{ cm} \quad BC = 15 \text{ cm} \quad \text{angle } ABC = 90^\circ$$

The point D lies on AB such that $DB = 9 \text{ cm}$ Work out the size of angle ACD

Give your answer correct to one decimal place.

o

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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No worked solution is saved yet.

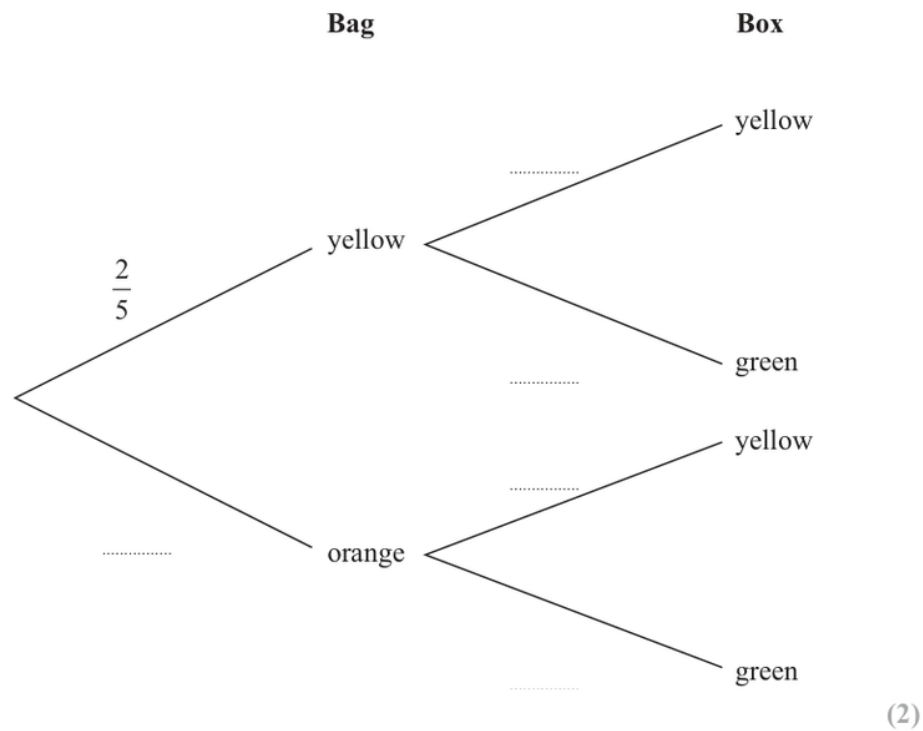
EA

PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 12** In a bag, there are only 2 yellow beads and 3 orange beads.
 In a box, there are only 3 yellow beads and 5 green beads.
 Chi takes at random a bead from the bag and a bead from the box.

(a) Complete the probability tree diagram.



(b) Work out the probability that Chi takes two beads of different colours.

.....
(3)**(Total for Question 12 is 5 marks)**

PAST PAPER SOLUTION

Q#: 12

Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 13****Marks: 6****TOPIC: Expanding Brackets**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

13 (a) Expand and simplify $7x(3x + 2)(2x - 5)$
(3)(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

 $y =$
(3)**(Total for Question 13 is 6 marks)**

EA

PAST PAPER SOLUTION

Q#: 13

Marks: 6

TOPIC: **Expanding Brackets**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 2****TOPIC: Circles, Arcs & Sectors**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

14 AOB is a sector of a circle with centre O and radius 12 cm

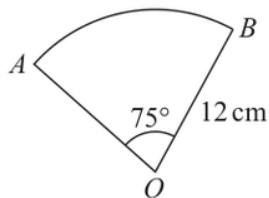


Diagram **NOT**
accurately drawn

Work out the area of the sector.

Give your answer correct to 3 significant figures.

..... cm²

(Total for Question 14 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 2

TOPIC: **Circles, Arcs & Sectors**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Algebraic Roots & Indices**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

15 $\frac{16^{x-3} \times 8^{x+1}}{32^x} = 2^n$

Express n in terms of x

Give your answer in its simplest form.

 $n = \dots\dots\dots$

(Total for Question 15 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 15

Marks: 3

TOPIC: **Algebraic Roots & Indices**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 16****Marks: 3**TOPIC: **Surds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

16 Show that $\frac{2\sqrt{7} + 2}{\sqrt{7} - 3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for Question 16 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 16

Marks: 3

TOPIC: **Surds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 17****Marks: 2**TOPIC: **Algebraic Proof**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

17 Use algebra to show that $0.9\dot{5}\dot{4} = \frac{21}{22}$

(Total for Question 17 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 17

Marks: 2

TOPIC: **Algebraic Proof**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

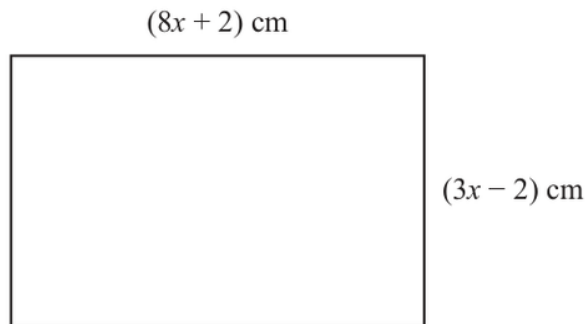
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

18 The diagram shows a rectangle.Diagram **NOT**
accurately drawnThe rectangle has length $(8x + 2) \text{ cm}$ and width $(3x - 2) \text{ cm}$ The area of the rectangle is 500 cm^2 (a) Show that $12x^2 - 5x - 252 = 0$

(1)

(b) Hence find the value of x

Give your answer correct to 3 significant figures.

 $x = \dots\dots\dots$

(3)

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18

Marks: 4

TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

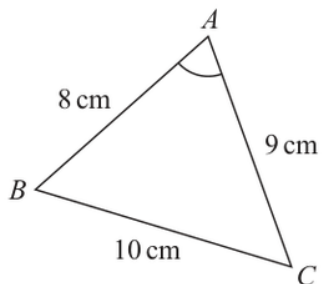
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

19 The diagram shows triangle ABC Diagram **NOT**
accurately drawn

$$AB = 8 \text{ cm} \quad BC = 10 \text{ cm} \quad CA = 9 \text{ cm}$$

Work out the size of angle BAC

Give your answer correct to one decimal place.

o

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19

Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION

Q#: 20 Marks: 3

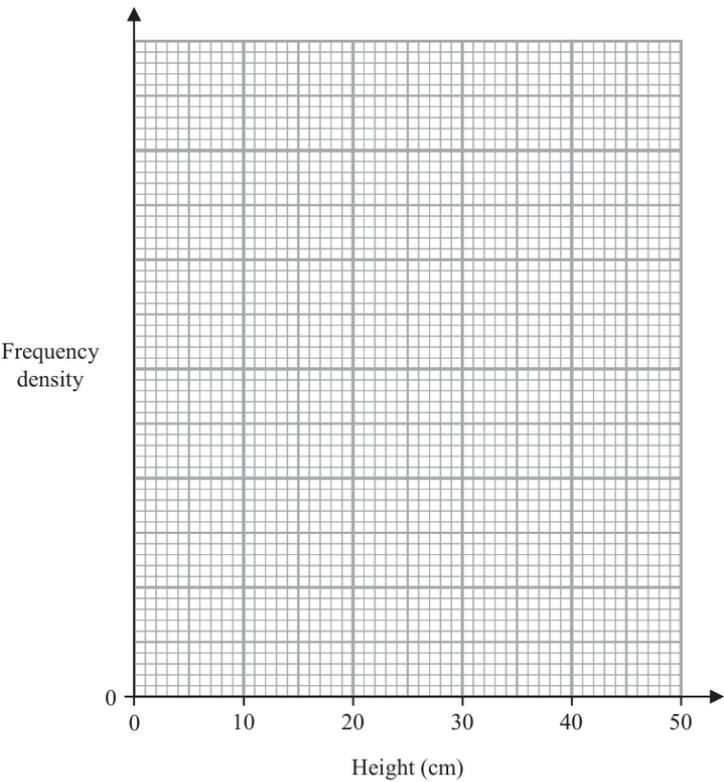
TOPIC: Histograms

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

20 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$0 < h \leq 5$	12
$5 < h \leq 15$	34
$15 < h \leq 30$	45
$30 < h \leq 50$	28

On the grid, draw a histogram for this information.
You must include a scale on the vertical axis.



(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Histograms**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION

Q#: 20 Marks: 3

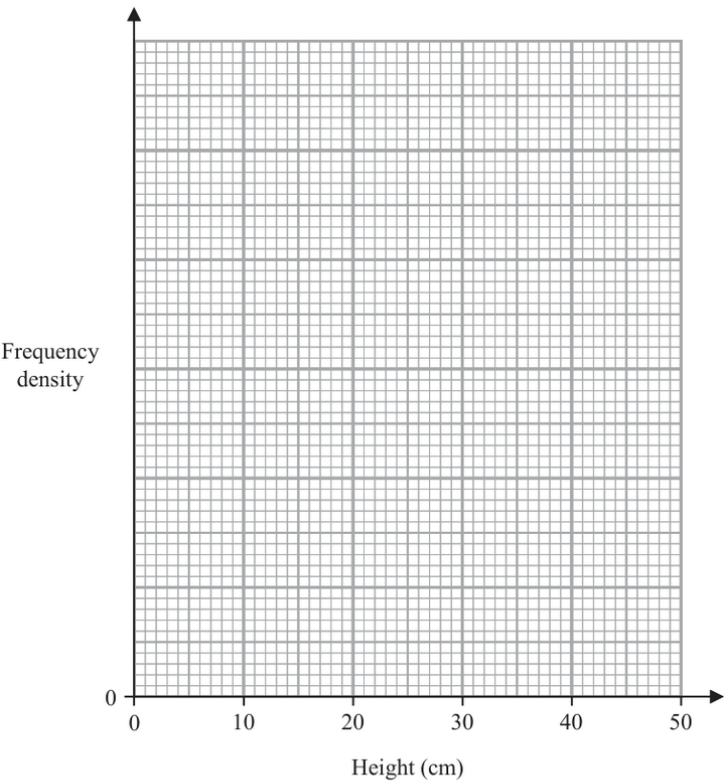
TOPIC: Histograms

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

20 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$0 < h \leq 5$	12
$5 < h \leq 15$	34
$15 < h \leq 30$	45
$30 < h \leq 50$	28

On the grid, draw a histogram for this information.
You must include a scale on the vertical axis.



(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Histograms**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

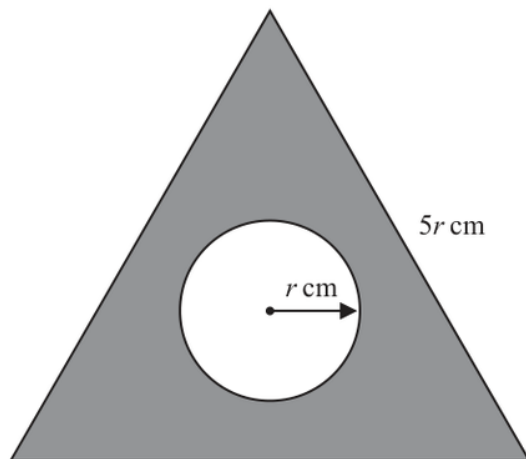
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

21 The diagram shows an equilateral triangle and a circle.Diagram **NOT**
accurately drawnThe equilateral triangle has sides of length $5r$ cmThe circle has radius r cmThe area of the shaded region is $610\pi \text{ cm}^2$ Work out the value of r

Give your answer correct to 3 significant figures.

 $r = \dots\dots\dots$ **(Total for Question 21 is 4 marks)**

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

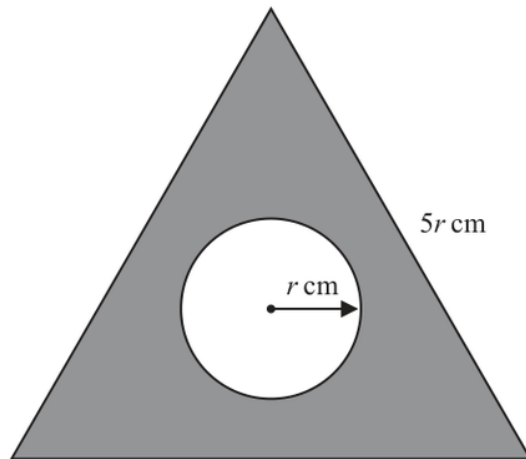
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

21 The diagram shows an equilateral triangle and a circle.Diagram **NOT**
accurately drawnThe equilateral triangle has sides of length $5r \text{ cm}$ The circle has radius $r \text{ cm}$ The area of the shaded region is $610\pi \text{ cm}^2$ Work out the value of r

Give your answer correct to 3 significant figures.

 $r = \dots\dots\dots$ **(Total for Question 21 is 4 marks)**

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 2****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

22 $9x^2 - 12x + q$ can be written in the form $(px - r)^2$ where p , q and r are integers.

Find the value of q

$q =$

(Total for Question 22 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 2

TOPIC: **Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 2****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

22 $9x^2 - 12x + q$ can be written in the form $(px - r)^2$ where p , q and r are integers.

Find the value of q

$q =$

(Total for Question 22 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 2

TOPIC: **Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

23 There are 15 buttons in a box.

3 of the buttons are red

2 of the buttons are pink

10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 23 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 23

Marks: 3

TOPIC: **Combined & Conditional Probability**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

23 There are 15 buttons in a box.

3 of the buttons are red

2 of the buttons are pink

10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 23 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 23

Marks: 3

TOPIC: **Combined & Conditional Probability**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 24****Marks: 3****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

24 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 24 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 3

TOPIC: **Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 24****Marks: 3****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

24 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 24 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 3

TOPIC: **Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 25****Marks: 3****TOPIC: Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

25 $a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m , n and p are integers.

Show your working clearly.

$a =$

(Total for Question 25 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 25

Marks: 3

TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 25****Marks: 3****TOPIC: Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

25 $a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m, n and p are integers.

Show your working clearly.

$a =$

(Total for Question 25 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 25

Marks: 3

TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

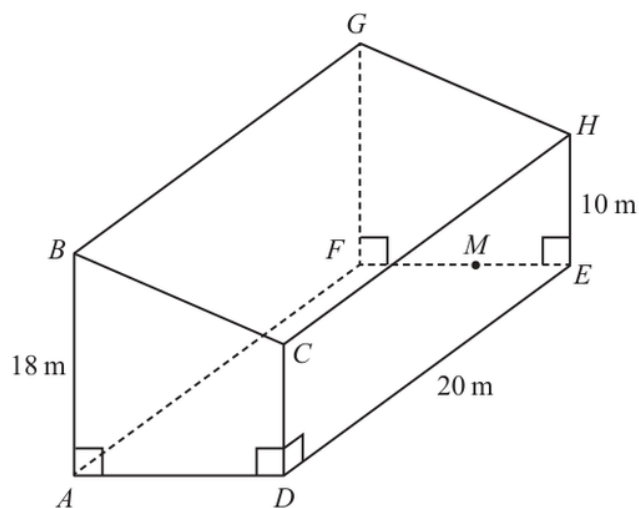
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 26** **Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

26 The diagram shows prism $ABCDEFGH$ Diagram **NOT**
accurately drawn

$$AB = FG = 18\text{ m} \quad DC = EH = 10\text{ m} \quad AD = FE$$

$$DE = CH = AF = BG = 20\text{ m}$$

$$\text{angle } BAD = \text{angle } ADC = \text{angle } CDE = 90^\circ$$

M is the point on FE such that $FM = \frac{3}{5}FE$

Given that angle $GMF = 60^\circ$

calculate the angle of elevation of G from D

Give your answer correct to one decimal place.

Show your working clearly.

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26

Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

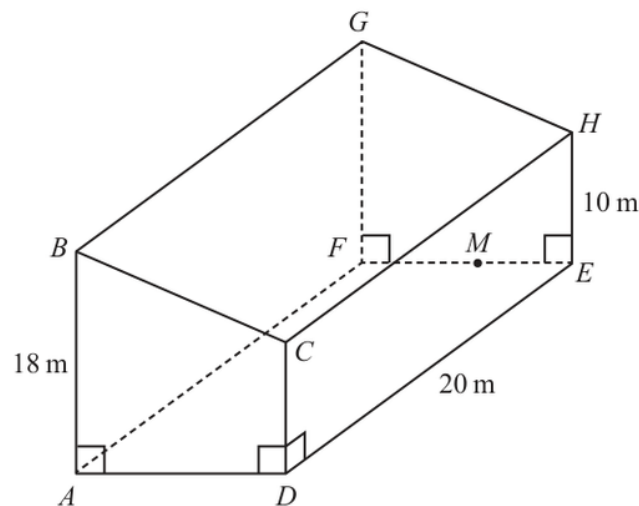
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 26****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

26 The diagram shows prism $ABCDEFGH$ Diagram **NOT**
accurately drawn

$$AB = FG = 18\text{ m} \quad DC = EH = 10\text{ m} \quad AD = FE$$

$$DE = CH = AF = BG = 20\text{ m}$$

$$\text{angle } BAD = \text{angle } ADC = \text{angle } CDE = 90^\circ$$

M is the point on FE such that $FM = \frac{3}{5}FE$

Given that angle $GMF = 60^\circ$

calculate the angle of elevation of G from D

Give your answer correct to one decimal place.

Show your working clearly.

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26

Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 27****Marks: 5****TOPIC: Coordinate Geometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

27 $ABCD$ is a rhombus with $AB = BC = CD = DA$

A has coordinates $(16, -9)$

C has coordinates $(24, 15)$

Find an equation of the line BD

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

Show your working clearly.

(Total for Question 27 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 27

Marks: 5

TOPIC: **Coordinate Geometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 27****Marks: 5****TOPIC: Coordinate Geometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

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PAST PAPER SOLUTION

Q#: 27

Marks: 5

TOPIC: **Coordinate Geometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 28****Marks: 4****TOPIC: Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

28 Solve $\frac{6x^3 + 5x^2 + x}{4x^2 - 1} \div \frac{15x^2 - x - 2}{8x - 4} = 2x$

Show clear algebraic working.

 $x = \dots\dots\dots$

(Total for Question 28 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 28

Marks: 4

TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 28****Marks: 4****TOPIC: Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

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Show clear algebraic working.

 $x = \dots\dots\dots$

(Total for Question 28 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 28

Marks: 4

TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA